

Effects of comprehensive lifestyle modification on blood pressure control: main results of the PREMIER clinical trial.



CONTEXT: Weight loss, sodium reduction, increased physical activity, and limited alcohol intake are established recommendations that reduce blood pressure (BP). The Dietary Approaches to Stop Hypertension (DASH) diet also lowers BP. To date, no trial has evaluated the effects of simultaneously implementing these lifestyle recommendations. OBJECTIVE: To determine the effect on BP of 2 multicomponent, behavioral interventions. DESIGN, SETTING, AND PARTICIPANTS: Randomized trial with enrollment at 4 clinical centers (January 2000-June 2001) among 810 adults (mean [SD] age, 50 [8.9] years; 62% women; 34% African American) with above-optimal BP, including stage 1 hypertension (120-159 mm Hg systolic and 80-95 mm Hg diastolic), and who were not taking antihypertensive medications. INTERVENTION: Participants were randomized to one of 3 intervention groups: (1) "established," a behavioral intervention that implemented established recommendations (n = 268); (2) "established plus DASH," which also implemented the DASH diet (n = 269); and (3) an "advice only" comparison group (n = 273). MAIN OUTCOME MEASURES: Blood pressure measurement and hypertension status at 6 months. RESULTS: Both behavioral interventions significantly reduced weight, improved fitness, and lowered sodium intake. The established plus DASH intervention also increased fruit, vegetable, and dairy intake. Across the groups, gradients in BP and hypertensive status were evident. After subtracting change in advice only, the mean net reduction in systolic BP was 3.7 mm Hg ($P < .001$) in the established group and 4.3 mm Hg ($P < .001$) in the established plus DASH group; the systolic BP difference between the established and established plus DASH groups was 0.6 mm Hg ($P = .43$). Compared with the baseline hypertension prevalence of 38%, the prevalence at 6 months was 26% in the advice only group, 17% in the established group ($P = .01$ compared with the advice only group), and 12% in the established plus DASH group ($P < .001$ compared with the advice only group; $P = .12$ compared with the established group). The prevalence of optimal BP (< 120 mm Hg systolic and < 80 mm Hg diastolic) was 19% in the advice only group, 30% in the established group ($P = .005$ compared with the advice only group), and 35% in the established plus DASH group ($P < .001$ compared with the advice only group; $P = .24$ compared with the established group). CONCLUSION: Individuals with above-optimal BP, including stage 1 hypertension, can make multiple lifestyle changes that lower BP and reduce their cardiovascular disease risk.